REPORT

"Smart College FAQ System: An NLP-Based Question Answering System Using Hybrid TF-IDF and Sentence-Embedding Retrieval"

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1. Introduction and Problem

Students ask the same routine questions every year about admissions, examinations, fees, the library, hostel accommodation, attendance, placements and scholarships. The answers exist, but they are spread across notices, department offices and static FAQ pages, so students must browse a list or visit an office, and staff answer the same questions repeatedly. Keyword search helps only when the student uses the same words as the stored text, and it returns the closest entry even when nothing relevant exists. This project builds the Smart College FAQ System, an end-to-end NLP retrieval application that accepts a question in ordinary English and returns the most relevant answer from a predefined college FAQ dataset, together with its category and a confidence value. Unrelated questions are rejected with a fallback message instead of a misleading answer.

Objectives: prepare a structured FAQ dataset; build a text-preprocessing pipeline and a TF-IDF baseline engine; integrate the pretrained sentence-embedding model all-MiniLM-L6-v2; combine both signals into a hybrid score with an intent-word boost and a confidence threshold; expose the engine through a FastAPI backend with SQLite history storage; deliver it through a Streamlit interface; and validate it with script-based tests at the preprocessing, retrieval and API levels.

2. Proposed System and Methodology

The system is retrieval-based: it never generates text, and every answer comes directly from data/faq_data.csv. The dataset holds 34 question–answer records grouped into 11 categories: Library (6), Hostel (6), Examination (4), Admission (3), Placement (3), Academic (3), Fees (2), Attendance (2), Scholarship (2), Student Activities (2) and General (1). The answers are general sample answers for demonstration, not verified college policies.

Each question passes through five preprocessing steps: lower-casing, removal of non-alphabetic characters, whitespace tokenisation, removal of NLTK English stop-words and WordNet lemmatisation (for example, "How can students register for placements?" becomes "student register placement"). The processed text is vectorised with TF-IDF (a 44-term vocabulary fitted on the 34 FAQ questions) for the lexical signal. The original question is encoded by all-MiniLM-L6-v2 into a 384-dimensional vector for the semantic signal, because the model benefits from the function words and word order that preprocessing removes. The model is used without fine-tuning, and the 34 × 384 FAQ embedding matrix is computed once at start-up. For FAQ record i, with TF-IDF cosine similarity T_i and semantic cosine similarity S_i, the final confidence is:

$$H_i = 0.30 \times T_i + 0.70 \times S_i; \quad C_i = \min(1, \max(0, H_i + B_i))$$

where B is a small boost of 0.05 added when an intent word (one of 18, such as hostel, register, scholarship or placement) occurs in both the question and the FAQ. The record with the highest score is selected; if that score is at least 0.50 its answer, category, matched question and confidence are returned, otherwise the system returns the category "Unknown" with a fallback message asking the student to contact the concerned department. Table 1 lists the retrieval parameters, which were set by the team and not tuned on a separate dataset.

Table 1: Retrieval parameters used in the project

Parameter	Value	Defined in
TF-IDF / semantic weights	0.30 / 0.70	hybrid_engine.py
Intent-word boost	0.05	hybrid_engine.py

Parameter	Value	Defined in
Hybrid acceptance threshold	0.50	hybrid_engine.py; main.py
TF-IDF baseline threshold	0.20	faq_engine.py
Embedding dimension	384	all-MiniLM-L6-v2

The system is a three-tier application (Figure 1). A Streamlit interface (app.py) sends the question as JSON to a FastAPI backend (backend/main.py), which validates it with a Pydantic model, calls the hybrid engine (loaded once at start-up) and stores every result in the question_history table of an SQLite database. The API provides five endpoints: /health, /ask, /faqs, /categories and /history, with Swagger documentation at /docs. The stack is Python 3.13, NLTK, scikit-learn, Sentence Transformers, pandas, FastAPI, Streamlit and SQLite, running on a CPU-only Windows machine with no model training.

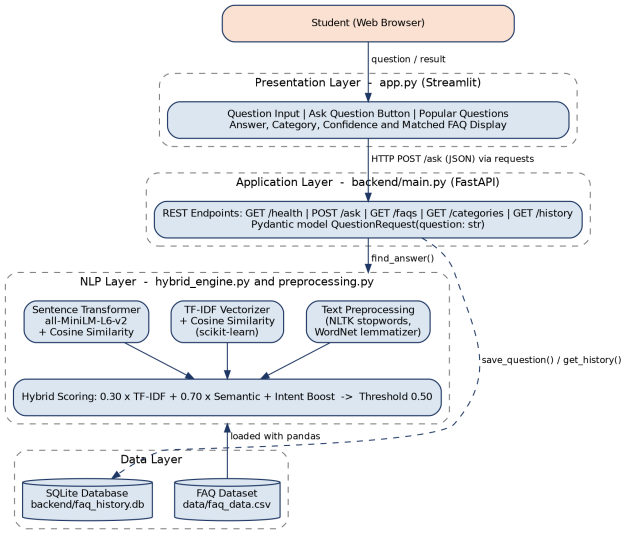


Figure 1: Layered architecture of the Smart College FAQ System

3. Results

The system was validated with test_api.py, which posts 14 questions to the running /ask endpoint and compares the returned category with the expected one: eleven college questions (one per category) and three unrelated questions that must return “Unknown”. The same questions were also run through the TF-IDF-only baseline to isolate the effect of the semantic component (Table 2).

Table 2: Validation set (n = 14): TF-IDF baseline vs hybrid confidence

No.	Question	Expected	TF-IDF	Hybrid	Result
1	What is the admission process?	Admission	100.00%	100.00%	PASS
2	Where can I find exam notices?	Examination	83.90%	96.68%	PASS
3	How can I pay my college fees?	Fees	100.00%	100.00%	PASS
4	Where do I get books?	Library	81.65%	83.45%	PASS
5	How can I get a hostel room?	Hostel	100.00%	100.00%	PASS
6	How much attendance do I need?	Attendance	70.71%	82.33%	PASS
7	I want to register for campus placement	Placement	100.00%	90.75%	PASS
8	How can I get a scholarship?	Scholarship	59.45%	88.50%	PASS
9	How can I contact my department?	Academic	100.00%	100.00%	PASS
10	How can I join a college club?	Student Activities	100.00%	100.00%	PASS
11	What are the college working hours?	General	100.00%	100.00%	PASS
12	What is the weather today?	Unknown	0.00%	11.29%	PASS
13	How do I cook rice?	Unknown	0.00%	15.85%	PASS
14	Tell me a joke.	Unknown	0.00%	10.33%	PASS

All 14 cases returned the expected category, giving 100% accuracy on this validation set (14/14). The eleven college questions scored between 82.33% and 100.00%, and the three unrelated questions between 10.33% and 15.85%; a manually tested variant, “What is the weather today? -”, scored 9.68%. The threshold of 0.50 therefore lies inside a gap of more than 66 percentage points. The six questions worded exactly as dataset questions score 100.00% because the sum is clipped at 1. For paraphrased questions the hybrid score is generally higher than the lexical one (“How can I get a scholarship?” rises from 59.45% to 88.50%, “How much attendance do I need?” from 70.71% to 82.33%), giving a wider margin above the threshold (Figure 2). The baseline also fails on meaning: for “Where can I see examination information?” it matched “How can I apply for an examination?” at 48.14% only through the shared word “examination”. The history database holds 42 stored records from interface, Swagger and test runs, confirming that logging works.

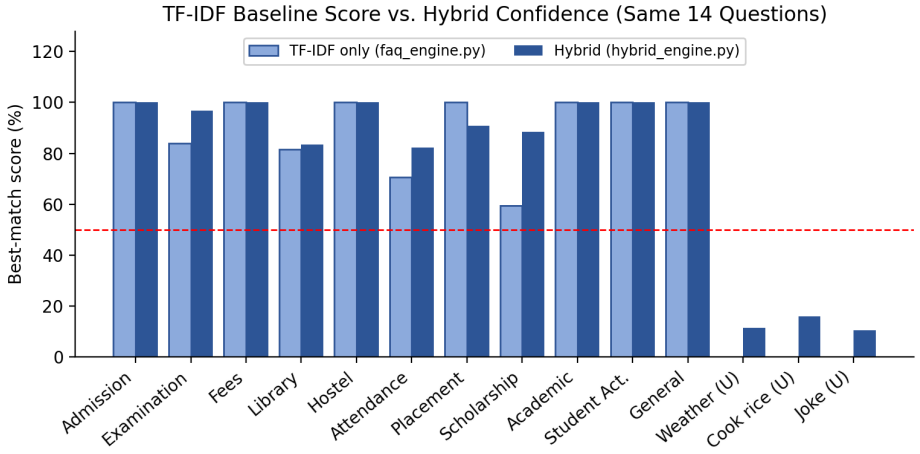


Figure 2: TF-IDF baseline score compared with hybrid confidence (same 14 questions)

Worked example. For the question “Where do I get books?”, which shares few words with the stored FAQ, the system returns the output in Table 3 and Figure 3. The implied semantic similarity is recovered from the reported scores using the hybrid equation, not measured directly. For the unrelated question “What is the weather today?”, none of its content words appear in the FAQ vocabulary, so the confidence of 11.29% comes only from the semantic branch and the answer is rejected.

Table 3: Platform output for the worked example

Output	Value
Matched FAQ	Where can I get books from the library?
Category / confidence	Library; 83.45% (above the 0.50 threshold)
Score decomposition	TF-IDF = 0.8165; implied semantic ≈ 0.842; boost = 0; hybrid = 0.8345
Answer returned	Students can get or borrow books from the college library according to the library rules and procedures.
Status	Relevant answer found; question stored in question_history

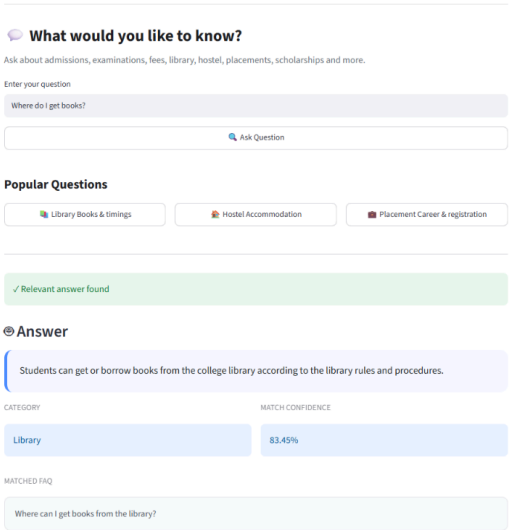


Figure 3: Streamlit result panel for the worked example

4. Discussion and Limitations

Combining a transparent lexical score with a pretrained semantic model lets common college questions be answered in different wordings without any training, and the confidence threshold keeps unrelated questions from receiving an answer. The results must be read with care. The 14-case validation set is small, was written by the project team, contains six questions worded exactly as in the dataset and checks only the category rather than the specific FAQ record. The TF-IDF baseline also returned the correct category for every case, so the observed benefit of the hybrid engine lies in stronger confidence margins on paraphrases, not in a higher accuracy. The result shows that the pipeline works as designed, not that 100% accuracy will be reached on real student questions, which contain spelling errors and partially related topics (for example, the canteen or transport) that were not tested and would probably receive intermediate scores.

Other limitations are: only 34 sample answers with uneven category coverage; English-only support and no spelling correction; weights, boost and threshold set manually; confidence clipped at 100%, which hides differences between exact matches; two hostel records with identical preprocessed text that TF-IDF cannot distinguish; lemmatisation without part-of-speech information; no authentication or admin interface and local-only deployment (127.0.0.1); and no formal measurement of response time or usability.

5. Conclusion and Future Scope

The project shows that a useful college FAQ assistant can be built from free, open-source components on an ordinary computer without training a model, while returning only pre-written answers and clearly stating when it cannot help. Combining TF-IDF and sentence-embedding similarity with an intent boost and a rejection threshold gives each answer a category, a confidence value and the matched FAQ, which makes the result transparent. Future work includes a larger, verified FAQ dataset and an independent validation set for tuning the parameters and checking the matched record, Hindi and Marathi support with a multilingual model, voice input and output, an admin interface for managing FAQs, re-ranking or fine-tuning of the embedding model, “Did you mean …?” suggestions from the top three candidates, authentication, cloud deployment and conversational follow-up questions.

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